Topology Optimization for Fracture Resistance using Differentiable Programming and Peridynamics

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Abstract

Fracture resistance is a critical consideration in engineering structures, particularly in applications where material failure can have catastrophic consequences. Traditional topology optimization methods often struggle to account for fracture propagation due to the inherently nonlocal nature of crack formation and growth. This work presents a framework for topology optimization that integrates bond-based peridynamics, a nonlocal continuum theory capable of modeling fracture without external failure criteria, with differentiable programming via JAX to enable gradient-based optimization of fracture-resistant designs. The peridynamic damage variable, defined as the volume-averaged fraction of broken bonds at each material point, is adopted as the optimization objective, providing a more direct indicator of discrete fracture than strain energy density or total fracture energy. Using this framework, optimized geometries are identified that provide highly localized reinforcement, forming impedance mismatches that reflect shock waves and suppress fracture. In one-dimensional examples, damage-based optimization produced fracture-resistant structures using substantially less material than strain energy density minimization and converged in less than 0.5% of the iterations required by the strain energy approach. These

results are extended to two dimensions, where consistent trends are observed. The combination of automatic differentiation, just-in-time compilation, and GPU acceleration makes damage-based peridynamic topology optimization computationally tractable, demonstrating the potential of this approach for designing robust, fracture-resistant structures under dynamic loading.

Keywords: Peridynamics, Fracture Resistance Topology Optimization, Automatic Differentiation

1 Introduction

Fracture resistance is a critical consideration in the design of engineering structures, particularly in applications where material failure can have catastrophic consequences [1]. Traditional topology optimization methods often struggle to account for fracture initiation and propagation due to the inherent difficulty in predicting crack nucleation and growth [2–5]. In classical continuum mechanics, material behavior is governed by spatial derivatives of the displacement field; however, when fractures or other discontinuities arise, these derivatives become undefined, requiring additional assumptions or external failure criteria to model crack evolution [1, 3, 6].

Peridynamics provides a unified framework that naturally overcomes these limitations [6–8]. In peridynamic theory, conservation of momentum is reformulated in integral form, allowing the analysis of problems involving displacement discontinuities without additional fracture criteria [7]. The continuum body is discretized into a collection of material points that interact not only with their nearest neighbors but with all points within a finite region known as the *horizon*. These interactions are governed by integral equations rather than spatial derivatives, and material points interact through "bonds" that describe their constitutive behavior and can progressively degrade and fail, allowing damage and fracture to emerge directly from the governing equations [6, 7, 9].

Beyond its ability to handle discontinuities, the nonlocal formulation of peridynamics offers two key advantages over classical continuum models. First, cracks and other discontinuities can form spontaneously as bonds break, without requiring predefined crack paths or enrichment strategies [6, 10]. Second, long-range interactions between material points are explicitly accounted for, enabling the modeling of force transmission over finite distances [7]. By resolving these long-range interactions, peridynamics captures small-scale dynamic phenomena, including dispersive wave propagation and stress redistribution, that are not represented in classical elasticity [9, 11]. Peridynamics therefore provides a natural framework for modeling crack initiation and propagation in one-, two-, and three-dimensional structures.

To date, these advantages have primarily motivated the use of peridynamics as an analysis tool for simulating fracture, damage, and dynamic failure, rather than as a framework for structural design [10–16]. However, the same features that distinguish peridynamics from classical continuum mechanics, namely its nonlocal formulation, intrinsic damage modeling, and ability to resolve long-range interactions, also make it well suited for design applications in which fracture resistance is a primary objective. This motivates the extension of peridynamics beyond analysis toward computational design methodologies.

As a computational design methodology, topology optimization enables systematic exploration of complex design spaces to identify structural configurations that outperform those derived from traditional engineering heuristics [17]. The method determines an optimal material layout within a prescribed design domain to achieve desired structural performance, and its relevance is further amplified by advances in additive manufacturing, which allow highly intricate optimized layouts to be fabricated with fidelity beyond the capabilities of traditional manufacturing processes [18].

Recent work has begun to explore the integration of peridynamics with topology optimization. Kefal et al. [19] first demonstrated that peridynamics could be combined with bi-directional evolutionary structural optimization (BESO) to optimize structures containing pre-existing cracks, leveraging the meshfree discretization of peridynamics to avoid the remeshing difficulties inherent to classical finite element approaches. Sohouli et al. [20] extended this to a continuous density-based formulation by applying the SIMP method within a peridynamic framework. Kendibilir et al. [21] further generalized these ideas to three-dimensional domains with surface cracks arising from additive manufacturing processes. More recently, Zhang et al. [22] proposed a peridynamics-based SIMP approach that constrains bond stretch, a uniquely peridynamic quantity, to guard against brittle fracture, and Feng et al. [23] introduced a multi-objective formulation that maximizes total fracture energy while penalizing early-time energy release, with sensitivities derived via a manually implemented adjoint method.

A common thread in the existing literature is the reliance on strain energy density or total fracture energy as the optimization objective [3, 23–25]. These quantities, while related to fracture, are inherently indirect indicators: strain energy density is a precursor to fracture that does not distinguish between localized damage and diffuse energy storage, and total fracture energy aggregates bond failures across the domain without discriminating between bonds that fail far from a crack path and those whose failure contributes to a propagating discrete crack. In peridynamic simulations, individual bonds can fail and release fracture energy in regions that never localize into a discrete crack; consequently, minimizing fracture energy or strain energy density does not necessarily target the suppression of discrete fracture. In this work, we instead adopt the peridynamic *damage* variable, defined as the volume-averaged fraction of broken bonds at each material point, as the optimization objective. Because damage accumulates only when a sufficient number of bonds within a node's family have

failed, elevated values of damage provide a localized and direct indicator of discrete fracture formation. In the standard peridynamic literature, a damage value of 0.5 corresponds to the loss of half the bonds in a node's family and is commonly associated with the formation of a crack surface [6, 10]. Optimizing against damage therefore targets the suppression of actual crack initiation rather than its energetic precursors, yielding fracture-resistant designs through highly localized reinforcement strategies that would not be identifiable using strain energy density–based methods [4, 26].

To enable gradient-based optimization of this damage-based objective, we implement the peridynamic model entirely within JAX, a fully differentiable array computing framework [27]. JAX provides automatic differentiation of arbitrary compositions of array operations and pure Python control flow, allowing gradients of the objective function with respect to design variables to be computed directly from the forward simulation code. This eliminates the need for manually derived adjoint equations, which become increasingly burdensome for nonlocal, history-dependent damage models, as well as finite-difference approximations that are prohibitively expensive for large systems. While automatic differentiation has been applied to topology optimization in finite element settings through frameworks such as AuTO [28] and JAX-FEM [29], the present work represents, to the authors' knowledge, the first application of a fully differentiable programming framework to peridynamic topology optimization. Combined with just-in-time compilation and GPU acceleration, this approach makes damage-based, gradient-driven peridynamic optimization computationally tractable at scales that would otherwise be prohibitive.

Using this computational framework, topology optimization is performed on the bond-based peridynamic model to directly suppress fracture. Traction boundary conditions are applied along the outer surface in a manner analogous to FEM, and the load is ramped to its full magnitude over a short duration using a smooth exponential function, generating shock waves that propagate through the structure [14, 30]. To prevent failure at the load application site, nodes within a no-damage zone are constrained such that their damage values remain zero. Because nodal velocities are computed at each material point, wave propagation can be explicitly tracked and visualized, allowing the optimizer to exploit dynamic loading effects inaccessible to traditional static or local methods. This proof-of-concept demonstrates that peridynamics, combined with gradient-based optimization enabled by automatic differentiation, can identify reinforcement strategies that suppress fracture in ways not achievable through strain energy density-based approaches alone.

2 The Peridynamic Formulation

This work serves as a proof of concept that peridynamic modeling can be used to perform topology optimization, and therefore employs a bond-based formulation. Extension to state-based peridynamics is a natural next step but exceeds the scope of the present study. In bond-based peridynamics, a material point $\mathbf{x}$ interacts with

all points $\mathbf{x}'$ within its horizon δ through elastic spring-like interactions. The peridynamic equation of motion, as presented by [6], is obtained by replacing the divergence of the stress tensor in the classical balance of linear momentum,

$$\rho\,\ddot{\mathbf{u}}(\mathbf{x},t) = \nabla\cdot\boldsymbol{\sigma}(\mathbf{x},t) + \mathbf{b}(\mathbf{x},t), \tag{1}$$

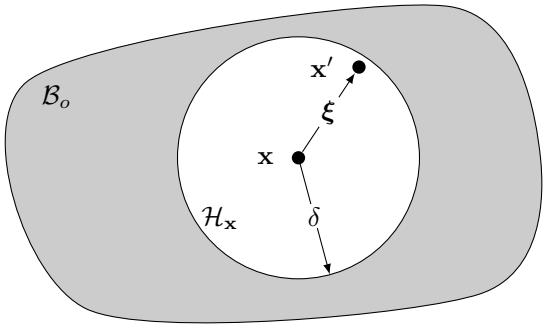


Fig. 1: Schematic of peridynamic body with a horizon

with an integral operator,

$$\rho\,\ddot{\mathbf{u}}(\mathbf{x},t) = \int_{\mathcal{H}_\mathbf{x}} \mathbf{f}(\mathbf{u}(\mathbf{x}',t)-\mathbf{u}(\mathbf{x},t),\mathbf{x}'-\mathbf{x})\;\mathrm{d}V_{\mathbf{x}'} + \mathbf{b}(\mathbf{x},t), \tag{2}$$

where $\mathbf{x}$ and $\mathbf{x}'$ are interacting material points, $\mathbf{u}(\mathbf{x},t)$ is the displacement field, and $\mathbf{f}$ is the pairwise force density function representing the force per unit volume squared exerted on point $\mathbf{x}$ by point $\mathbf{x}'$ [6]. Body forces are represented by $\mathbf{b}(\mathbf{x},t)$ and ρ is the mass density. The pairwise force function and body forces are integrated over the horizon $\mathcal{H}_\mathbf{x}$ surrounding the point. As is common in peridynamic formulations, the relative position of two points in the reference configuration is denoted

$$\boldsymbol{\xi} = \mathbf{x}' - \mathbf{x}, \tag{3}$$

and their relative displacement is denoted

$$\boldsymbol{\eta} = \mathbf{u}(\mathbf{x}',t) - \mathbf{u}(\mathbf{x},t). \tag{4}$$

The horizon $\mathcal{H}_\mathbf{x}$ is defined as a spherical neighborhood of radius δ centered at point $\mathbf{x}$, comprising all points whose interactions collectively influence the response at $\mathbf{x}$. Points outside the horizon do not interact with $\mathbf{x}$, so that

$$|\boldsymbol{\xi}| > \delta \;\Rightarrow\; \mathbf{f}(\boldsymbol{\eta},\boldsymbol{\xi}) = 0 \quad \forall\,\boldsymbol{\eta}. \tag{5}$$

Damage in peridynamics is most naturally implemented through bond breakage [6, 10]. The scalar bond stretch s is defined as

$$s = \frac{|\boldsymbol{\xi} + \boldsymbol{\eta}| - |\boldsymbol{\xi}|}{|\boldsymbol{\xi}|}, \tag{6}$$

and when s exceeds a critical value s_0, the bond can no longer sustain any force and is irreversibly broken [6]. The critical stretch s_0 is given by

$$s_0 = \sqrt{\frac{5G_0}{9k\delta}}, \tag{7}$$

where G_0 is the critical energy release rate, k is the bulk modulus, and δ is the horizon radius. The integrity of each bond is tracked through a history-dependent influence state μ, defined as

$$\mu(t, \boldsymbol{\xi}) = \begin{cases} 1, & \text{if } s(t', \boldsymbol{\xi}) < s_0 \quad \text{for all } 0 \le t' \le t, \\ 0, & \text{otherwise,} \end{cases} \tag{8}$$

where $\mu = 1$ indicates an intact bond and $\mu = 0$ indicates a broken bond. The influence state is incorporated into the constitutive model through the vector-valued pairwise force density

$$\mathbf{f}(\boldsymbol{\eta}, \boldsymbol{\xi}, t) = c\, s(t, \boldsymbol{\xi})\, \mu(t, \boldsymbol{\xi})\, \frac{\boldsymbol{\xi} + \boldsymbol{\eta}}{|\boldsymbol{\xi} + \boldsymbol{\eta}|}, \tag{9}$$

where the scalar magnitude $cs\mu$ determines the force intensity and the unit vector $(\boldsymbol{\xi}+\boldsymbol{\eta})/|\boldsymbol{\xi}+\boldsymbol{\eta}|$ defines the force direction along the deformed bond. The micromodulus c (spring constant) is given for a spherical horizon by

$$c = \frac{18k}{\pi\delta^4}, \tag{10}$$

where k is the bulk modulus. A well-known consequence of the bond-based formulation is that the Poisson ratio is constrained to $\nu = 1/4$ in three dimensions and $\nu = 1/3$ in two dimensions [7]. This limitation can be removed by adopting a state-based formulation.

The damage at a material point $\mathbf{x}$ is a post-processed quantity obtained by computing the volume-averaged fraction of broken bonds within its family. Nodal damage $\phi(\mathbf{x}, t)$ is defined as [6]

$$\phi(\mathbf{x}, t) = 1 - \frac{\int_{\mathcal{H}_\mathbf{x}} \mu(\mathbf{x}, t, \boldsymbol{\xi})\, \mathrm{d}V_{\boldsymbol{\xi}}}{\int_{\mathcal{H}_\mathbf{x}} \mathrm{d}V_{\boldsymbol{\xi}}}, \tag{11}$$

where $\phi = 0$ corresponds to a material point with all bonds intact and $\phi = 1$ corresponds to a point that has lost all of its bonds. Broken bonds cannot sustain tensile

loading; therefore, when a bond fails, its load is redistributed to the remaining intact bonds in the neighborhood. This redistribution drives a progressive accumulation of damage that can eventually culminate in fracture. The failure of a single bond does not immediately trigger catastrophic failure or crack initiation; instead, damage emerges only after a sufficient number of bonds within a node's family have broken, reflecting the inherently nonlocal nature of peridynamic interactions. This damage-accumulation mechanism is a distinguishing feature of peridynamics that local continuum models cannot replicate [6, 8, 10, 11]. Leveraging this capability to resolve highly localized damage enables targeted structural reinforcement at critical locations and provides insight into the conditions under which cracks or catastrophic failure are likely to occur.

3 Performing Topology Optimization Using Peridynamics

A peridynamic model was constructed to perform topology optimization on a one-dimensional bar loaded in tension, with a surface traction of 500 MPa applied to each end via a smooth exponential ramp function of the form $F(t) = F_0(1-e^{-\beta(t-t_0)})$ that increases rapidly to the prescribed magnitude and then asymptotically approaches a constant value. This load magnitude was selected because it produces a clean fracture in a bar with uniform thickness. The material properties of AISI 304 stainless steel were used throughout [31], with Young's modulus $E = 193$ GPa, bulk modulus $k = 161$ GPa, density $\rho = 8000$ kg/m^3, and critical energy release rate $G_0 = 12,000$ J/m^2.

The design variable in the one-dimensional model is the nodal thickness $\tau(\mathbf{x})$, which scales the nodal volume $V(\mathbf{x})$ used in the discrete force integration. Because the peridynamic internal force at a material point is computed by summing pairwise force contributions weighted by the neighbor volumes, increasing thickness at a node proportionally increases both the force it exerts on its neighbors and the force it receives, thereby increasing local stiffness. The lumped mass at each node is also scaled by the nodal volume, so the design variable simultaneously modifies both stiffness and inertia, which in turn affects local wave speeds and impedance characteristics. Thickness is bounded between 0.3 and 100.0 (dimensionless, relative to the initial uniform thickness of 1.0), and these bounds are enforced by projecting the design variables after each gradient update.

As a baseline, nodal thickness was first optimized to minimize strain energy density, a quantity commonly used in classical topology optimization to improve fracture resistance. The L^∞ norm was selected for the strain energy objective because it targets the worst-case (maximum) strain energy density across all nodes, which is most

directly related to incipient fracture at critical locations. The objective function is

$$\begin{aligned} \min_{\tau} \quad & \left\| \tfrac{1}{2} \int_{\mathcal{H}_{\mathbf{x}}} w(\boldsymbol{\xi})\, V(\boldsymbol{\xi})\, \mathrm{d}\boldsymbol{\xi} \right\|_{L^\infty} \\ \text{subject to} \quad & 0.3 \leq \tau(\mathbf{x}) \leq 100.0, \end{aligned} \tag{12}$$

where $w(\boldsymbol{\xi})$ is the micropotential (strain energy density per unit volume associated with each bond), $V(\boldsymbol{\xi})$ is the volume associated with the neighboring point, and $\tau(\mathbf{x})$ is the nodal thickness at point $\mathbf{x}$. All objective functions are evaluated at the final simulation time T, after the dynamic loading has been fully applied and the resulting damage has evolved. To accelerate convergence and prevent overflow, the strain energy density was divided by a normalization factor of 1.0×10^{12}, chosen to bring the objective to order unity for the initial configuration. Thickness constraints were imposed with additive manufacturing in mind, ensuring printability of the final structure.

The optimization procedure was then applied to adjust the thickness distribution to minimize damage directly. In contrast to the strain energy baseline, the L^1 norm was selected for the damage objective because it promotes sparse solutions, concentrating reinforcement at a small number of critical locations rather than distributing material broadly. The loss function is

$$\begin{aligned} \min_{\tau} \quad & \|\phi(\mathbf{x}, T)\|_{L^1} \\ \text{subject to} \quad & 0.3 \leq \tau(\mathbf{x}) \leq 100.0, \end{aligned} \tag{13}$$

where $\phi(\mathbf{x}, T)$ is the nodal damage at the final simulation time T, as defined in Equation 11. To further explore the sensitivity of the optimization to the choice of norm, damage was also minimized using the L^2 norm,

$$\begin{aligned} \min_{\tau} \quad & \|\phi(\mathbf{x}, T)\|_{L^2} \\ \text{subject to} \quad & 0.3 \leq \tau(\mathbf{x}) \leq 100.0. \end{aligned} \tag{14}$$

An initial element spacing of $\Delta x = 0.25$ m was used over a bar of length $L = 10$ m with a horizon radius of $\delta = 0.9$ m. To examine how the optimal thickness distribution evolves with spatial resolution, the discretization was progressively refined by reducing the element length while keeping the horizon fixed, and the optimization was repeated at each resolution. The simulation and optimization parameters are summarized in Table 1.

Table 1: Simulation and optimization parameters.

Parameter	**Symbol**	**Value**
Material properties (AISI 304)		
Young's modulus	E	193 GPa
Bulk modulus	k	161 GPa
Mass density	ρ	8000 kg/m^3
Critical energy release rate	G_0	12,000 J/m^2
Geometry and discretization		
Bar length	L	10 m
Element spacing (coarse)	Δx	0.25 m
Element spacing (fine)	Δx	0.17 m
Horizon radius	δ	0.9 m
Loading		
Surface traction magnitude	F_0	500 MPa
Load profile		Exponential ramp
Time integration		
Scheme		Velocity-Verlet
Time step	Δt	5.0×10^{-8} s
Optimization		
Optimizer		Adam
Learning rate	α	1.0
Momentum parameters	β_1, β_2	0.9, 0.999
Thickness bounds (1D)	τ	$[0.3, 100.0]$
Density bounds (2D)	ρ_d	$[0, 1]$
Damage termination threshold	$\phi_{\max}$	0.5

4 Implementation

The optimization framework was implemented in Python using the JAX library, which provides high-performance numerical computing through just-in-time (JIT) compilation and automatic differentiation [27]. JAX offers a largely one-to-one correspondence with the NumPy API while enabling execution on CPUs, GPUs, or TPUs, allowing the array-based computations central to peridynamic simulations to be accelerated without restructuring the underlying code.

4.1 Time Integration

The peridynamic equation of motion (Equation 2) is integrated in time using the explicit Velocity-Verlet scheme, which provides second-order accuracy and is well suited for the explicit dynamics considered here. Given the current displacement $\mathbf{u}^n$, velocity $\mathbf{v}^n$, and acceleration $\mathbf{a}^n$ at time step n, the update proceeds as

$$\mathbf{u}^{n+1} = \mathbf{u}^n + \mathbf{v}^n \Delta t + \tfrac{1}{2}\mathbf{a}^n (\Delta t)^2, \qquad \mathbf{v}^{n+1} = \mathbf{v}^n + \tfrac{1}{2}(\mathbf{a}^n + \mathbf{a}^{n+1})\Delta t, \tag{15}$$

where $\mathbf{a}^n = \mathbf{F}^n/\rho$ is the acceleration at the current configuration, $\mathbf{a}^{n+1}$ is computed from forces at the updated positions, and $\Delta t = 5.0 \times 10^{-8}$ s is the fixed time step. This time step was selected empirically to ensure numerical stability for the given horizon, grid spacing, and material properties. The simulation is advanced for a total of $N = T/\Delta t$ time steps, where T is the total simulation time, and the objective function is evaluated at the final time step.

4.2 Automatic Differentiation and Bond Breakage

A key advantage of JAX is its support for automatic differentiation (AD) of arbitrary compositions of array operations and pure Python control flow, enabling gradients of scalar objective functions to be computed directly from the simulation code. This eliminates the need for finite-difference approximations or manually derived adjoint equations, both of which become prohibitively expensive and error-prone for large nonlocal systems.

A central implementation detail is the treatment of bond breakage under automatic differentiation. The influence state μ (Equation 8) is a discontinuous function that transitions from 1 to 0 when the bond stretch exceeds the critical value s_0. In the implementation, this conditional is realized using JAX's `jnp.where` operation, which selects between two branches based on the stretch criterion. Under JAX's AD, `jnp.where` propagates gradients only from the selected branch: for intact bonds ($s < s_0$), the gradient of the force with respect to the design variables is computed via the standard chain rule, while for broken bonds ($s \geq s_0$), the gradient contribution is zero. The resulting algorithmic derivatives are not classical gradients of the non-smooth objective but rather generalized derivatives consistent with the piecewise-smooth structure of the forward model. The optimizer therefore receives gradient information exclusively from bonds that remain intact or are near the breaking threshold, and the optimization effectively performs a form of subgradient descent. To ensure numerical stability, broken bonds are assigned a small residual influence state value of $\varepsilon = 10^{-10}$ rather than exactly zero, preventing division-by-zero errors in downstream computations. Despite the piecewise nature of the damage model, the Adam optimizer converges reliably, as the gradient signal from near-threshold bonds provides sufficient information to guide the design toward configurations that prevent bond failure. A smooth surrogate for bond degradation (e.g., a sigmoid approximation of the influence state) could in principle provide classical differentiability but was not needed for convergence in the present study.

4.3 Optimization

Optimization was performed using Optax, a gradient-based optimization library designed for compatibility with JAX's automatic differentiation pipeline [32]. The

Adam optimizer was employed with a learning rate of $\alpha = 1.0$ and momentum parameters $\beta_1 = 0.9$, $\beta_2 = 0.999$. The relatively large learning rate was found to produce efficient convergence for this problem; the adaptive gradient scaling of Adam dynamically adjusts step sizes based on first- and second-moment estimates of the gradients. This adaptive behavior is especially beneficial for the highly nonlinear peridynamic damage model, where gradient magnitudes can vary substantially across the domain and throughout the optimization process. Box constraints on the design variables are enforced by projecting onto the feasible set after each gradient update. For all optimization studies, simulations were advanced until the damage at every node within the optimizable region fell below a value of 0.5, ensuring that no crack surface forms in the final design. The source code for the implementation is publicly available at https://github.com/johntfoster/pdjax.

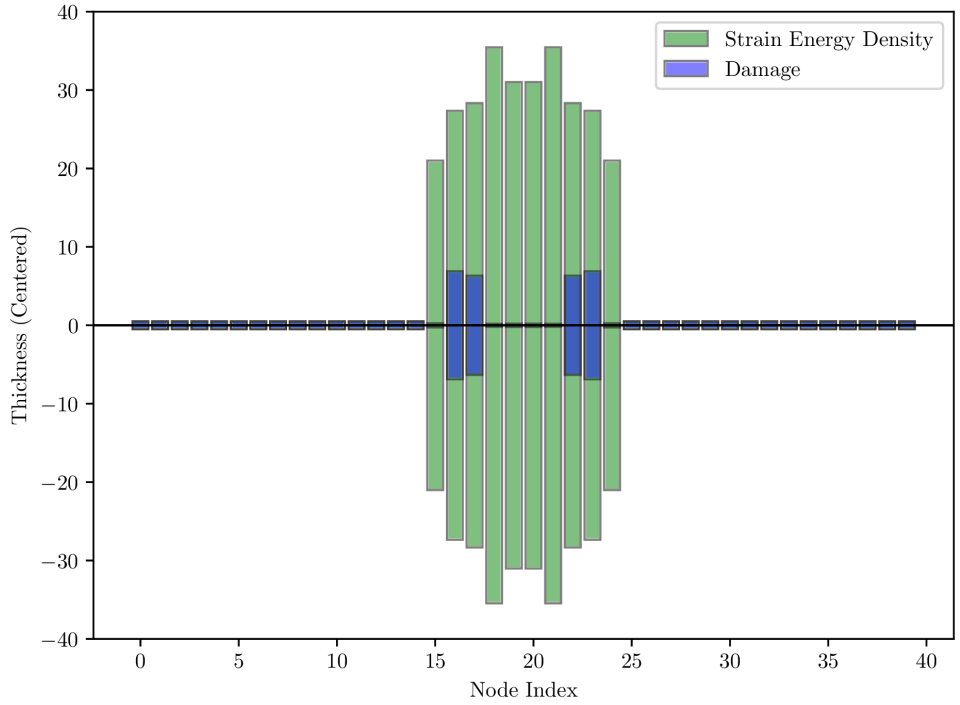


Fig. 2: Optimized nodal thickness distributions for the 1D bar obtained by minimizing strain energy density (green) and the L^1 norm of damage (red). The damage-optimized profile achieves fracture suppression with substantially less added material than the strain energy density-optimized profile.

5 1D Results

5.1 Minimizing Strain Energy Density

A peridynamic framework was first used to perform topology optimization of a one-dimensional bar by optimizing the nodal thickness distribution to minimize strain energy density, providing a baseline for comparison with damage-based optimization. The resulting optimized thickness profile is shown in green in Figure 2, with

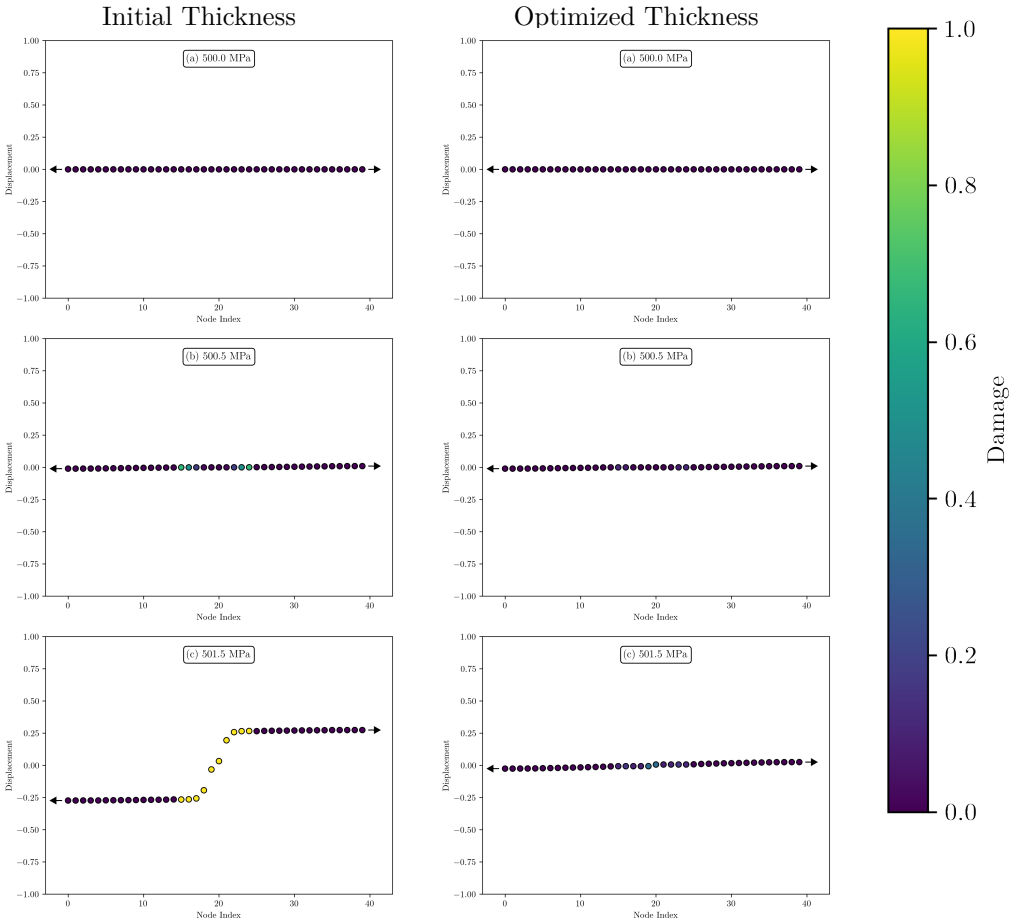


Fig. 3: Damage evolution in the 1D bar at three successive load levels. The left column shows the bar with initial uniform thickness, and the right column shows the bar with strain energy density-optimized thickness. Damage (colorbar) accumulates progressively in the uniform bar, leading to fracture, whereas the optimized thickness distribution suppresses damage below the critical threshold.

initial thickness values of 1.0 everywhere. Comparison of the damage evolution in the optimized and initial thickness configurations, shown in Figure 3, demonstrates that minimizing strain energy density reduces the number of damaged nodes in the structure.

In this work, a damage value of 0.5 is taken to indicate a fully damaged node, corresponding to the point at which half of the initial bonds connected to a node have failed and force transmission is no longer possible, allowing a crack to initiate. To prevent fracture at the loading boundaries, a no-damage region was imposed on the outer 15 nodes at each end of the bar. Within this region, bond breakage is suppressed by preventing the influence state from transitioning to zero regardless of the stretch, and the thickness is fixed at 1.0. The remaining 10 nodes outside these regions were treated as optimizable. In the initial configuration, all 10 of these nodes exceeded the 0.5 damage threshold, whereas no nodes exceeded this threshold when the bar with the optimized thickness profile was loaded.

Because stress is defined as force per unit area, increasing thickness in regions experiencing damage provides an intuitive means of reducing structural damage. Accordingly, the optimized thickness profile obtained by minimizing strain energy density is physically consistent with basic mechanical principles. However, such solutions largely reflect conventional engineering intuition and do not exploit the full capabilities of peridynamics. In contrast, the damage-based optimization approach introduced next exploits the nonlocal, dynamic nature of peridynamics to uncover fracture-mitigating designs that cannot be obtained from strain energy density minimization alone.

Both the strain energy density and damage-based optimization approaches were subject to the same termination criterion described in Section 4 (damage below 0.5 at all optimizable nodes), but their convergence behavior differed markedly. When strain energy density was minimized, the optimization required 4,659 steps to reduce damage below the threshold across all optimizable nodes. In contrast, when the L^1 norm of damage was minimized, the optimization converged in only 18 steps. This substantial difference in convergence rate further underscores the effectiveness of directly optimizing damage within the peridynamic framework.

5.2 Minimizing Damage

Thickness was then optimized to minimize damage in the 1D bar. The resulting optimized thickness profile, shown in red in Figure 2, exhibits increased thickness in only 4 of the 10 optimizable nodes, concentrated at two targeted locations. All damage in the bar was prevented by reinforcing only these locations. Minimizing damage eliminated fracture while using approximately 90% less material than the geometry optimized for strain energy density, as evidenced by the maximum thickness in the damage-optimized profile being less than 30% of the maximum obtained when optimizing for strain energy density. By directly modeling damage evolution under dynamic loading,

peridynamics enables fracture to be suppressed through highly localized reinforcement at critical wave-interaction sites, an outcome that minimizing strain energy density, which lacks explicit damage and wave-tracking capability, cannot achieve.

In the damage-optimized thickness profile, the locations of maximum thickness do not coincide with the regions where fracture occurs in the bar with the initial thickness. Rather than increasing thickness at the central nodes where fracture occurs under the initial configuration (node indices 20 and 21), the optimizer increases thickness at locations two nodes away from the center (node indices 17, 18, 23, and 24). The bond-based nature of peridynamics provides insight into this behavior.

In a peridynamics simulation, applying a surface traction to the bar generates a shock wave that propagates through the structure. Because peridynamics computes interactions through bonds, nodal velocities are evaluated throughout the domain, allowing the resulting velocity field to be visualized and the propagation of shock waves to be observed directly. Visualization of velocity wave propagation in the bar with a damage-optimized thickness profile reveals why relatively small thickness increases-compared to those obtained from strain energy density optimization-are sufficient to prevent fracture as shown in Figure 4.

The velocity waves propagate through the bar and constructively interfere, producing the greatest loads at nodes 17, 18, 23, and 24. The nonlocal formulation of peridynamics captures these long-range wave interactions. In solid mechanics, abrupt changes in mechanical impedance at material interfaces can reflect incident waves and inhibit smooth transmission of displacement or force; such interfaces are referred to as impedance mismatches [33]. The thick, off-center sections in the optimized geometry act as impedance mismatches, reflecting shock waves and mitigating damage within the bar. When shock-wave effects are included, the optimizer therefore increases thickness at locations of constructive wave interference, reducing overall damage and producing more localized reinforcement than designs optimized using strain energy density alone.

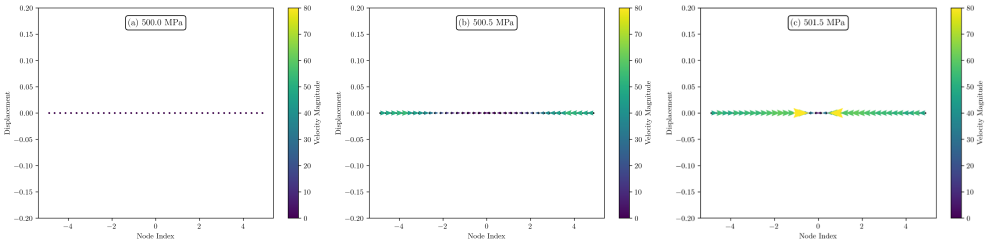


Fig. 4: Velocity wave propagation in the 1D bar with damage-optimized thickness at three successive time steps. Shock waves initiated at the boundaries propagate inward and constructively interfere at the locations of increased thickness, which act as impedance mismatches that reflect the waves and suppress fracture.

The increased thickness at these locations forms an impedance mismatch that protects the central portion of the bar from damage by blocking shock-wave propagation. By preventing wave transmission and subsequent constructive interference, the optimized design suppresses fracture in the bar.

To assess the sensitivity of damage-based optimization to the choice of loss function, the L^2 norm of damage was also minimized and compared to results obtained using the L^1 norm. Because a uniform initial thickness combined with problem symmetry can inhibit symmetry-breaking during optimization, random initial thickness values between 0.3 and 5.0 were assigned to each node, and simulations were repeated using different random seeds. The optimized thickness distributions, shown in Figure 5 obtained using the L^2 norm exhibited variability depending on the initial thickness.

Across the simulations, two distinct trends in the optimized geometries emerged. In three of the five cases, the optimizer produced a single region of increased thickness at the central nodes where fracture occurs in the uniformly thick bar. In the remaining two cases, the optimized thickness increased at the same off-center locations identified in the L^1 damage-optimized solution. Notably, the optimized profile obtained from one initialization exhibited both trends simultaneously. However, only the L^2-optimized geometry that closely resembled the L^1 solution-corresponding to the off-center reinforcement-was able to completely eliminate damage in the bar. Among the remaining designs, only one partially reduced damage, while the others failed to prevent fracture.

These results further highlight the importance of creating an impedance mismatch to suppress damage accumulation under dynamic loading. The differing behavior of the L1 and L2 norms provides insight into why the L^1 norm more reliably identifies this effective design. When used in optimization, the L^2 norm tends to reduce the loss by distributing changes across many locations, leading to more diffuse thickness modifications. In contrast, the L^1 norm promotes sparse solutions by driving thickness changes to zero at most locations, enabling the optimizer to concentrate material at a small number of critical nodes. This sparsity allows the L^1-based optimization to more consistently form localized impedance mismatches that block shock-wave propagation and prevent fracture.

Refinement of the discretization increases the spatial resolution within the bar, allowing velocity wave propagation to be resolved more accurately and the resulting structural reinforcement to be more precisely distributed. In the coarsest model, the optimized thickness field reaches its maximum at four discrete locations. As the discretization is refined, these four reinforced regions persist while the number of nodes comprising each region increases (Figure 6). Finer discretizations also capture higher-frequency wave components that contribute to the overall dynamic response, leading to additional localized thickness variations within each reinforced region. Although the primary locations of reinforcement remain unchanged as the element length is reduced, the thickness distribution becomes increasingly spatially resolved, with elevated thickness distributed over consecutive nodes rather than concentrated at a

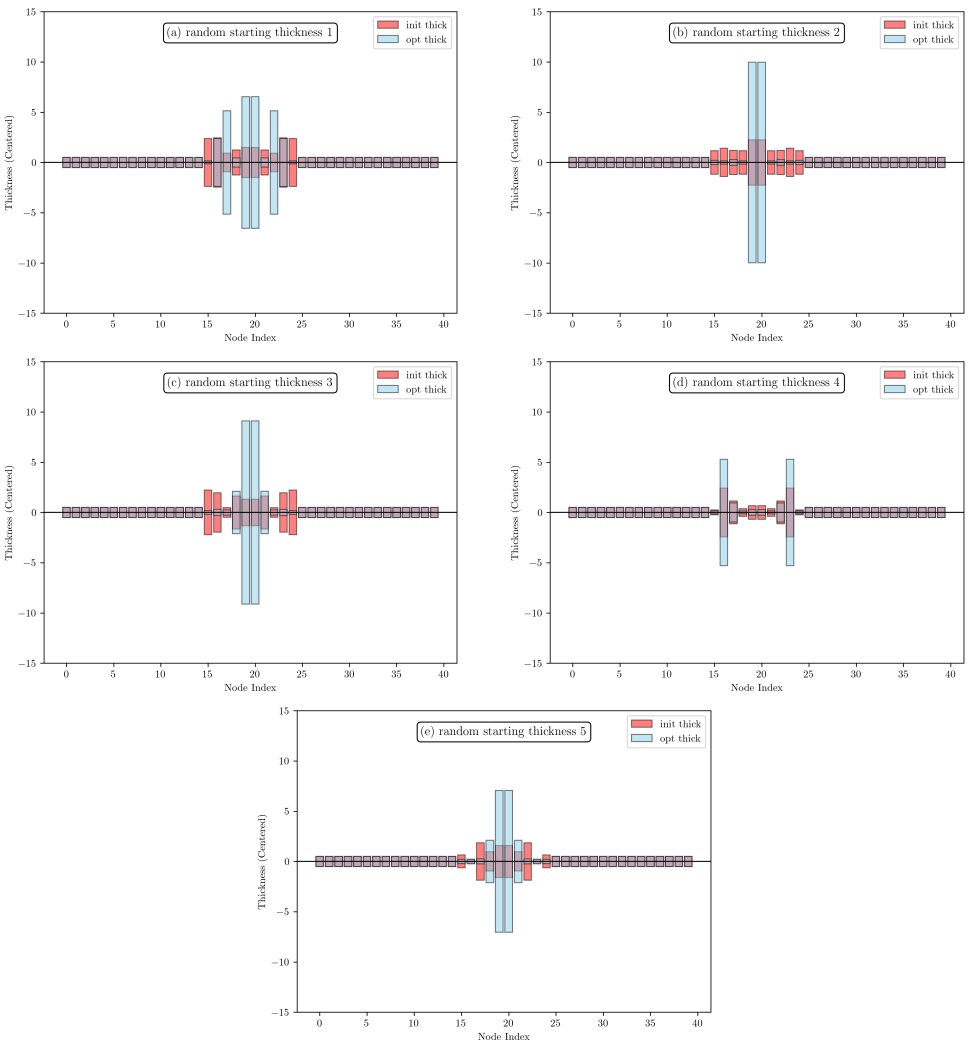


Fig. 5: Effect of initial thickness on L^2-norm damage optimization. The initial thickness is shown in red with the optimized thickness overlaid in blue. Of the five initializations, two distinct geometric trends emerged: (a, b, c, e) thickness concentrated at the center, and (d) off-center reinforcement resembling the L^1-optimized profile. Only the off-center pattern (d) prevented damage entirely.

single point. Consistent qualitative trends in the optimized geometry are observed, but strict convergence of the fine-scale design features is not expected, as continued refinement allows progressively shorter-wavelength dynamics to influence the optimization. The resulting refined distribution forms localized impedance mismatches

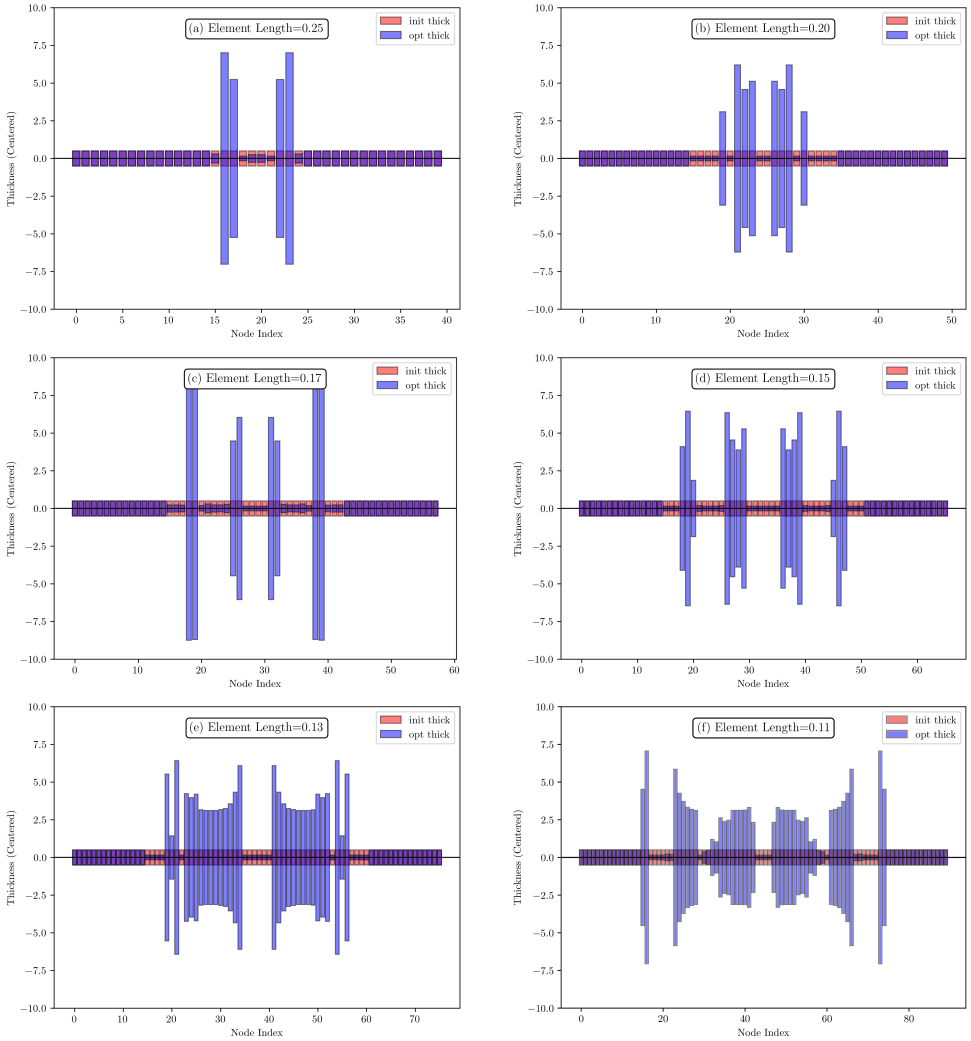


Fig. 6: The optimized thickness distribution in the 1D bar is shown as the element length is reduced from 0.25 to 0.17. As the element length is reduced, the locations of structural reinforcement remain unchanged, while the thickness field becomes increasingly spatially resolved. Reinforced regions that appear as discrete maxima in the coarsest model evolve into broader zones spanning multiple nodes with finer discretization, forming localized impedance mismatches that more effectively limit wave propagation and reduce damage.

that more effectively limit shock-wave propagation, thereby reducing damage in the bar and increasing its fracture resistance.

6 2D Results

6.1 Minimizing Strain Energy Density

The model was extended to two dimensions and the topology optimization procedure was repeated. In 2D, the design variable at each node is a density $\rho_d(\mathbf{x}) \in [0, 1]$, where 0 indicates a void and 1 indicates solid material. As with thickness in 1D, the density scales the nodal volume used in the discrete force integration, so that nodes with $\rho_d = 0$ contribute no force to the system. In principle, obtaining a binary (solid/void) design requires integer programming, since structures composed of intermediate-density material are not physically realizable. In this work, a continuous density relaxation is employed in which the optimization is carried out over the continuous interval $[0, 1]$ and the resulting density field is subsequently thresholded at $\rho_d = 0.5$ to obtain a binary structure. This approach is feasible because peridynamics, unlike finite element methods, can accommodate zero-density regions throughout the optimization without introducing stiffness-matrix singularities, which penalization methods such as SIMP are typically used to mitigate in FEM-based formulations.

The optimized topology obtained by minimizing strain energy density is shown in Figure 7. As has been well documented in the literature, a truss-like structure emerges from the density-based optimization. Damage was effectively prevented in the optimized structure.

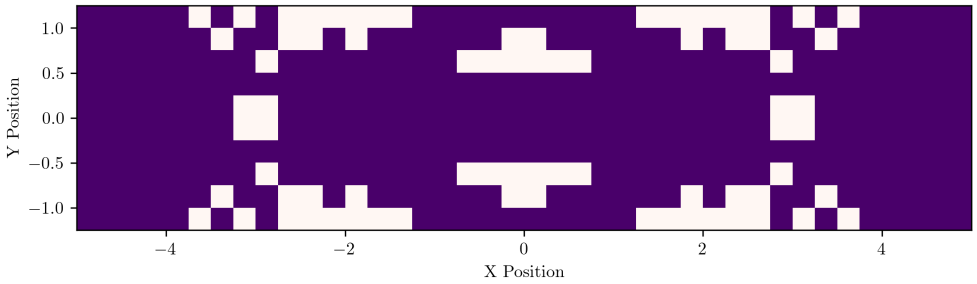


Fig. 7: The optimized 2D structure obtained by minimizing strain energy density. Consistent with prior literature, a truss-like topology emerges from the density-based optimization.

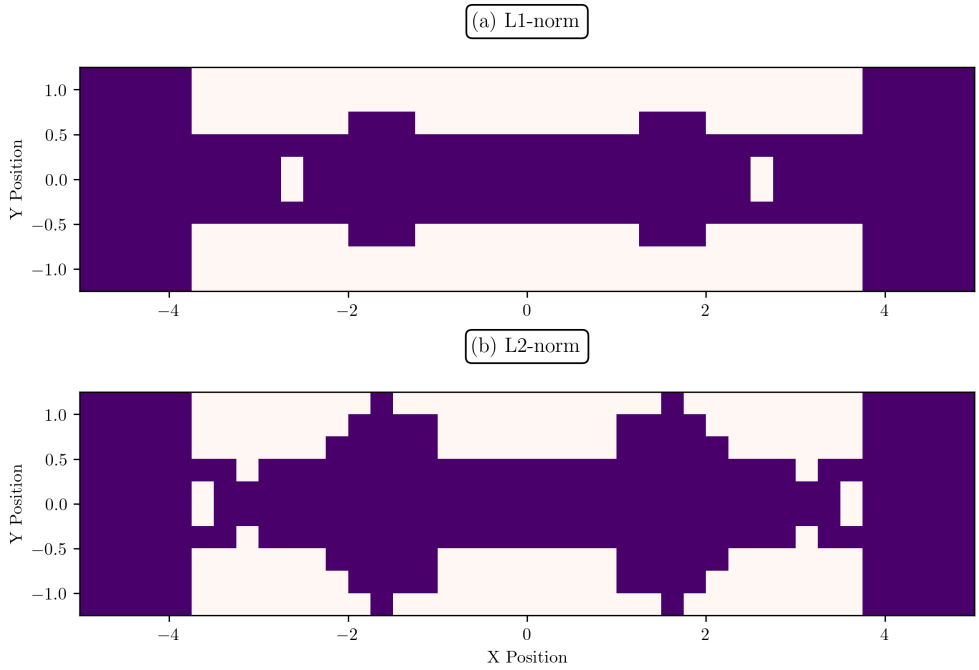


Fig. 8: The optimized geometries obtained by minimizing the L^1 (a) and L2 (b) norms of damage are shown. Consistent with the 1D results, both cases exhibit their maximum thickness at approximately 45% of the bar length, which is effective in creating an impedance mismatch that suppresses damage propagation.

6.2 Minimizing Damage

Density was then optimized to minimize damage in the 2D bar. To assess the sensitivity of damage-based optimization to the choice of norm, damage was minimized using the L^1 and L^2 norms, and the resulting structures were compared. The optimized geometries obtained by minimizing the L^1 and L^2 norms of damage are shown in Figure 8. Despite slight differences between the two, both damage-optimized designs eliminate fracture while requiring substantially less material than the strain energy density-optimized geometry, consistent with the trends observed in 1D. In 1D, the strain energy density-optimized geometry prevented damage but required more than $10\times$ as much material as the damage-optimized geometry, and this trend persisted in 2D. Of the 300 nodes outside the no-damage region, 224 had a density of 1 in the strain energy density-optimized geometry, compared to only 128 in the L^1-norm and 164 in the L^2-norm damage-optimized models. Optimizing for damage therefore produced a geometry that required approximately 40% less material than when strain energy density was minimized, further highlighting the advantages of using peridynamic damage as the optimization objective.

The optimized topologies exhibit distinct geometric features for the two optimization methods; however, in both cases the regions of greatest thickness are located approximately 45% along the length of the bar, consistent with trends observed in the 1D topology optimization results. As in 1D, clear differences emerge between the L^1- and L^2-norm-based optimizations. In the L^1-optimized geometry (Figure 8a), damage is mitigated by concentrating material within highly localized regions, whereas in the L^2-optimized geometry (Figure 8b), damage suppression is achieved by distributing material over a broader region. Despite these differences, both designs successfully prevent fracture, indicating that each optimized thickness distribution introduces effective impedance mismatches that restrict shock-wave propagation.

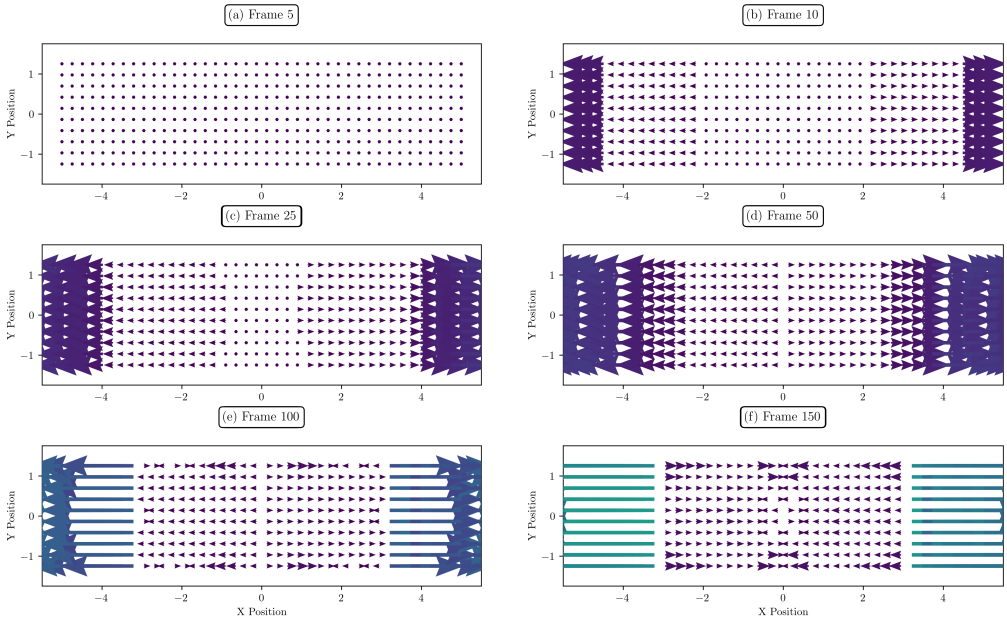


Fig. 9: Velocity wave propagation in the original uniform-thickness 2D structure. Arrow length is proportional to the local velocity magnitude. In the absence of an impedance mismatch, velocity waves generated at the boundaries propagate freely through the structure and undergo repeated reflections (b, c), leading to constructive interference near the exterior regions (d, e). This wave amplification concentrates velocity and stress within the outer third of the structure, promoting damage accumulation, indicated by the lighter-colored velocity arrows, and ultimately fracture initiation (f). The uniform structure lacks localized reinforcement to disrupt wave transmission, allowing damaging wave interactions to persist throughout the domain.

The role of impedance mismatch in 2D is further clarified by visualizing the velocity field, where the magnitude of the velocity waves is indicated by the arrow length. As in the 1D case, velocity waves generated by the applied traction propagate nonlocally

through the structure. In the original, fully solid bar, these impulse waves travel from the boundaries toward the interior, reflect, and constructively interfere just outside the center of the bar, producing localized regions of high dynamic loading-evidenced by large velocity arrows-that lead to damage accumulation and fracture (see Figure 9). In contrast, the optimized 2D designs introduce regions of increased thickness at these locations of constructive interference. These thickened regions act as impedance mismatches, reflecting incident waves and preventing their transmission toward the center of the structure, thereby suppressing damage (see Figure 10).

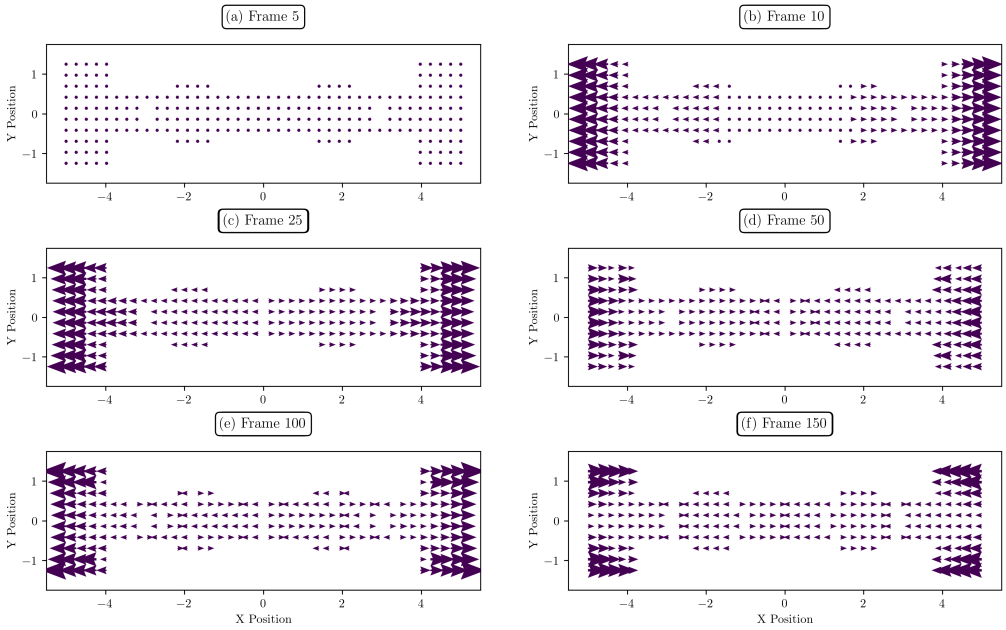


Fig. 10: Velocity wave propagation in the 2D structure optimized to minimize the L^1 norm of damage with an element length of 0.25. Arrow length is proportional to the local velocity magnitude. Waves initiate at the outer boundaries (a), propagate inward (b), and subsequently reflect outward (c), where constructive interference amplifies the velocity magnitude. The impedance mismatch induced by the optimized thickness distribution is evident, as the largest velocities are confined to the thickened outer regions of the structure (e, f), while wave transmission into the interior is suppressed.

As the discretization is refined in 2D (shown in Figure 11), the optimized designs diverge more noticeably than in 1D, as increasingly higher-frequency wave components are resolved and incorporated into the optimization. Consequently, strict convergence of the fine-scale topology is not observed; continued refinement allows shorter-wavelength dynamics to influence the optimized geometry. Nevertheless, consistent structural themes emerge across all discretizations. As in 1D, the optimized designs consistently feature regions of increased thickness offset from the center of

the bar. While the precise size and location of these regions vary slightly with discretization, in all cases they occur outside the midspan and correspond to locations of elevated dynamic loading.

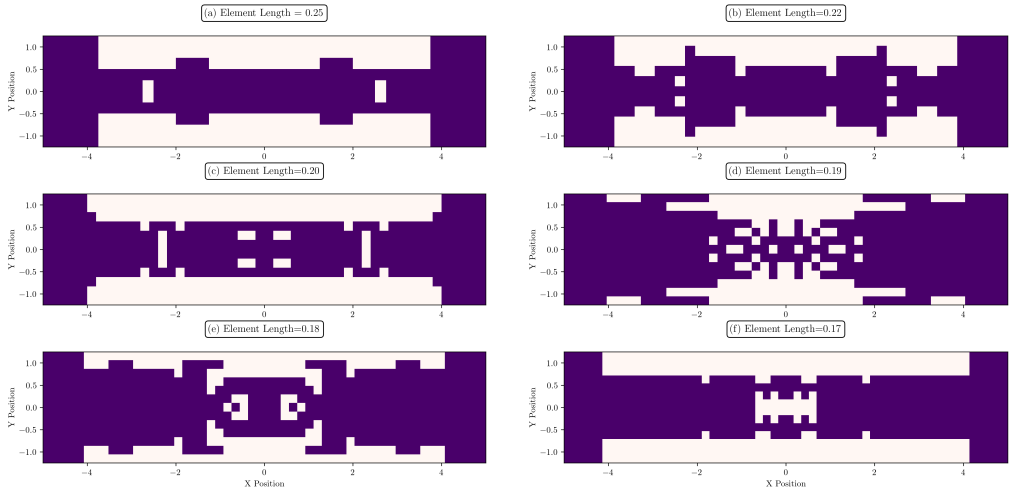


Fig. 11: The optimized density distribution in the 2D bar as the element length is reduced from 0.25 to 0.17. As the discretization is refined, the primary locations of structural reinforcement persist while the topology becomes increasingly complex, with additional geometric features emerging to exploit higher-frequency wave dynamics. Thickened regions outside the center form impedance mismatches, and open regions near the center further inhibit wave transmission.

In addition to these thickened regions, mesh refinement reveals increased geometric complexity near the center of the bar. For the coarsest discretization (element length 0.25), the optimized geometry remains relatively uniform through the midsection. As the discretization becomes finer, the topology in and around the center evolves to include regions of reduced material or open space. The combination of increased thickness outside the center and reduced material within the center enhances the impedance mismatch across the structure, further inhibiting wave transmission and constructive interference.

These effects are most clearly observed when visualizing the velocity field for the optimized geometry obtained using an element length of 0.17, as shown in Figure 12. In this case, higher-frequency shock waves are clearly resolved, and the influence of the optimized topology on wave propagation becomes evident. A concentrated band of material appears at approximately node positions $x \approx \pm 3.5$, mirroring the thickened regions observed in the 1D optimized design and in the coarser 2D discretizations. The presence of this thick band produces a pronounced reduction in velocity magnitude downstream, directly illustrating the formation of an impedance mismatch. Additional bracket-like features near $x \approx \pm 2$ introduce secondary impedance mismatches.

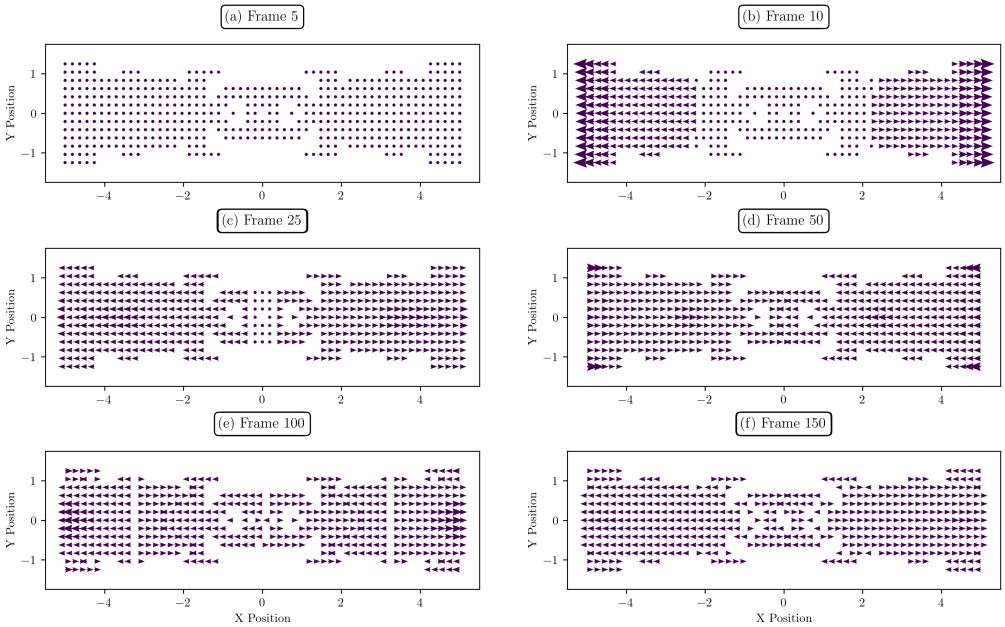


Fig. 12: Velocity wave propagation in the 2D structure optimized to minimize the L^1 norm of damage with an element length of 0.17. Arrow size and length are proportional to the local velocity magnitude. Waves initiate at the outer boundaries (a), propagate inward (b), and reflect outward (c), where constructive interference amplifies the velocity magnitude. The impedance mismatch induced by the optimized thickness distribution confines the largest velocities to the thickened outer regions (e, f), while wave transmission into the interior is suppressed.

Together with the open region at the center of the bar, these features prevent velocity amplification, as evidenced by the nearly constant velocity magnitude within the bracketed region and minimal large arrows.

Figure 13 compares the damage evolution in a fully solid bar with that of a bar incorporating the optimized topology. In the optimized structure, no nodes reach full damage, whereas the fully solid bar exhibits extensive regions of complete failure. This comparison demonstrates that the optimized thickness distributions suppress constructive wave interference and prevent fracture, an outcome enabled by the nonlocal formulation of peridynamics.

6.3 Limitations

Several limitations of the present proof-of-concept study should be noted. First, only tensile loading of a single material (AISI 304 stainless steel) is considered; the sensitivity of the optimized designs to other loading scenarios, load magnitudes, and material parameters has not been explored. Second, the optimization formulation imposes only box constraints on the design variables and does not include an explicit volume or mass constraint, which is standard practice in topology optimization. While the damage-based objective naturally converges to material-efficient solutions, a volume constraint would enable a more rigorous comparison of material usage across different objective functions. Third, the bond-based peridynamic formulation constrains the Poisson ratio, limiting quantitative accuracy for materials with Poisson ratios far from the bond-based values. Fourth, the 2D density-optimized designs are post-processed via thresholding at $\rho_d = 0.5$, but the thresholded designs have not been re-simulated to verify that the damage criterion is still satisfied. Finally, no density filtering or regularization is applied, which may lead to mesh-dependent features in the optimized topologies. These limitations define clear directions for future work, including extension to state-based peridynamics, incorporation of volume constraints, validation against experimental data, and application to more complex geometries and loading conditions.

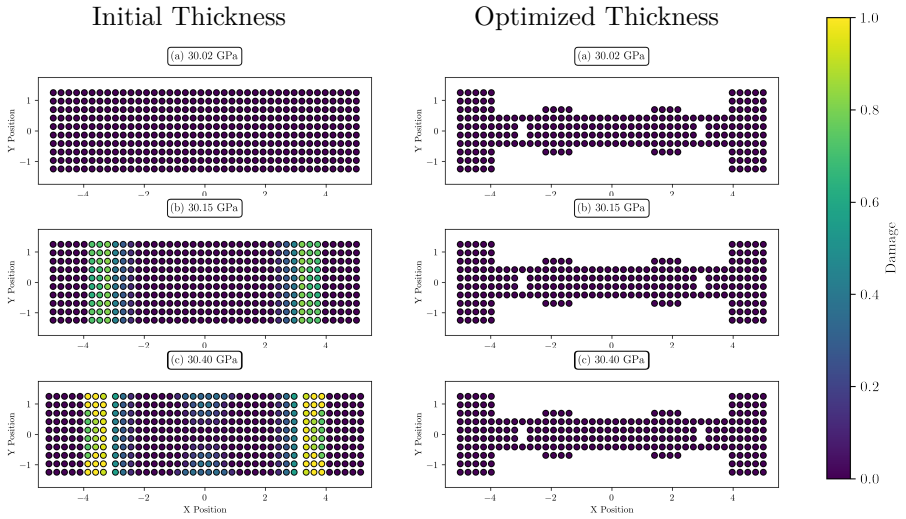


Fig. 13: The deformation of the initial bar in 2D with uniform thickness $t = 1.0$ is shown on the left, while the deformation of the optimized bar is shown on the right. Whereas the optimized structure suppresses damage growth, the original structure exhibits increasing damage accumulation with increasing applied load.

7 Conclusion

This work demonstrates that peridynamics-based damage modeling enables the design of highly fracture-resistant topologies by explicitly resolving long-range wave interactions under dynamic loading. The nonlocal formulation captures shock-wave propagation and nodal damage evolution, revealing that fracture is most effectively suppressed by targeted, highly localized reinforcement placed slightly off center, where constructive wave interference produces peak loads. These thickness increases form impedance mismatches that reflect incident waves and prevent fracture initiation, a mechanism consistently observed when optimizing both the L^1 and L^2 norms of damage in one- and two-dimensional systems. In contrast, strain energy density minimization, which lacks explicit damage and wave-tracking capability, requires substantially more material and cannot achieve the same level of fracture suppression. Critically, because the peridynamic damage variable provides a localized and direct indicator of discrete fracture, with a value of 0.5 corresponding to the formation of a crack surface, optimizing against damage targets the suppression of actual fracture rather than its energetic precursors. Incorporating automatic differentiation within a JAX-based implementation enables efficient gradient evaluation and GPU acceleration, overcoming the traditionally prohibitive computational cost of peridynamic sensitivity analysis and making large-scale, damage-informed topology optimization tractable without the need for manually derived adjoint equations. Together, these results establish peridynamics as a powerful and scalable framework for wave-informed structural optimization, providing insights for designing fracture-resistant structures that are inaccessible to conventional strain energy-based methods. Extension to state-based peridynamics and three-dimensional domains represents a natural direction for future work.

Supplementary information.

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Declarations

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- **Conflict of interest:** The authors declare no conflict of interest.
- **Ethics approval:** Not applicable.
- **Consent for publication:** Not applicable.
- **Data availability:** The datasets generated during this study are available from the corresponding author on reasonable request.
- **Code availability:** The source code for the peridynamic topology optimization framework is publicly available at https://github.com/johntfoster/pdjax.
- **Author contributions:** M.A. Schmitz: Conceptualization, Methodology, Software, Investigation, Writing – Original Draft. J.T. Foster: Conceptualization, Supervision,

Writing – Review & Editing. J.E. Fowler: Conceptualization, Writing – Review & Editing. All authors reviewed and approved the final manuscript.

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